

SECTION B

Answer **all** questions in the spaces provided.

8. (a) Iron(III) oxide, Fe_2O_3 , and carbon monoxide react to give iron and carbon dioxide.

Give the equation for this reaction. Explain why it is described as a redox process. [2]

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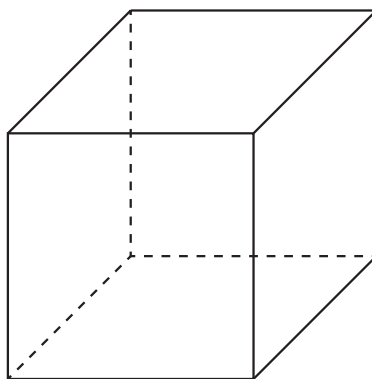
- (b) The iron obtained from the blast furnace contains some sulfur as an impurity. This is removed during Basic Oxygen Steelmaking (BOS), by adding magnesium, which removes the sulfur as magnesium sulfide, MgS .

350 tonnes of impure iron used for BOS contain 0.02 % of sulfur. Calculate the mass, in **kilograms**, of magnesium needed to remove all the sulfur. [2]

Mass = kg

- (c) Magnesium sulfide has the same crystal structure as sodium chloride.

Use the diagram below to show the crystal structure of magnesium sulfide, clearly labelling the formula of each species present. [2]



- (d) Magnesium sulfide reacts with water producing gaseous hydrogen sulfide and magnesium hydroxide.



- (i) State what would be seen during the reaction apart from gas bubbles. [1]

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- (ii) A student added a small sample of magnesium sulfide of mass 0.224 g to some water in a fume cupboard.

Calculate the maximum volume, in cm^3 , of hydrogen sulfide produced at 25°C . [3]

Volume = cm^3

- (e) When barium sulfide is added to water a similar reaction occurs to that described in part (d) with gaseous hydrogen sulfide and barium hydroxide being the products.

State with a reason, how the observation for this reaction would be different from that seen with magnesium sulfide. [2]

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- (f) Barium hydroxide can also be produced by adding barium oxide to water.

Give the equation for this reaction and estimate the pH of the product. [2]

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(g) Explain why it is unlikely that stable compounds containing Ba^{3+} ions can exist.

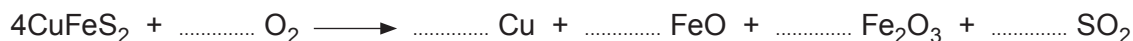
[2]

Examiner
only

16



8. (a) Copper can be extracted from the mineral chalcopyrite. The mineral is smelted by heating with air. The equation for this reaction is as follows.



- (i) Balance the equation above. [1]
- (ii) Give the full electronic configuration of a copper atom. [1]

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- (iii) A sample of rock contains 1.30 % by mass of chalcopyrite. Assuming this is the only source of copper in the rock, calculate the percentage by mass of copper in the sample. [2]

Percentage = %

- (iv) During the smelting Cu_2S is produced. This is collected and blown with air to produce copper.



This is classified as a redox reaction. Use oxidation numbers to explain which elements have been oxidised and which reduced. [3]

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- (b) Sulfur dioxide can react with oxygen and water to form sulfuric acid. Although sulfuric acid is a strong acid, it does not have to be a concentrated acid.

Explain the difference between the terms *strong acid* and *concentrated acid*. [2]

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- (c) (i) A student was given an aqueous solution of an acid HX and was asked to find out if it was a strong or weak acid. 25.0 cm^3 of the acid HX required 15.90 cm^3 of $0.0125\text{ mol dm}^{-3}$ sodium hydroxide solution for complete neutralisation.

Calculate the concentration of the acid. Assume that HX reacts with the sodium hydroxide in a 1:1 molar ratio. [2]

Concentration = mol dm^{-3}

- (ii) A teacher measured the pH of the aqueous solution and found it to be 2.10. She told the student that the acid in the solution must be a strong one. Is she correct? Justify your answer. [2]

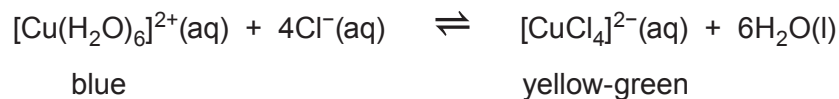
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- (d) The commonest complex ion that copper forms in solution is the hexaaquacopper(II) ion, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$. If concentrated hydrochloric acid is added to a solution containing hexaaquacopper(II) ions a new complex forms as the following equilibrium is established.



The forward reaction is endothermic.

- (i) State and explain what you would observe on adding water to the equilibrium mixture. [2]

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- (ii) State and explain what you would observe on heating the equilibrium mixture. [2]

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8. (a) The diagram below shows part of the atomic emission spectrum of hydrogen.



- (i) Use the letter **A** to label the line of longest wavelength on the diagram. [1]
- (ii) Explain why hydrogen atoms emit only certain definite frequencies of visible light. [2]

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- (b) The ionisation energy of a hydrogen atom is 2.18×10^{-21} kJ.

- (i) Explain what this statement means. [2]

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- (ii) Calculate the minimum frequency of radiation required to ionise a hydrogen atom in its ground state. [3]

Frequency = s^{-1}



- (c) Hydrazine is a compound of hydrogen and nitrogen only. It is a colourless, flammable liquid which was used in various rocket fuels.

0.160 g of hydrazine on vaporisation at 398 K and 1 atm pressure has a volume of 163 cm³.

Calculate its volume at 273 K and 1 atm pressure and hence show that its molecular formula is N₂H₄. [3]

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- (d) (i) Draw a dot and cross diagram to show the electron arrangement in hydrazine, N₂H₄. Show outer electrons only. [2]

- (ii) Hydrazine contains polar covalent bonds between nitrogen and hydrogen atoms. State what is meant by a *polar* covalent bond. [1]

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- (e) Hydrazine acts as a base in a similar way to ammonia.

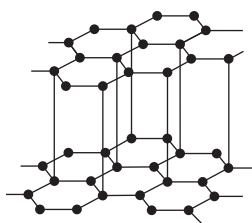
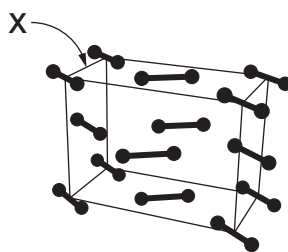
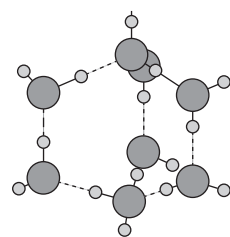
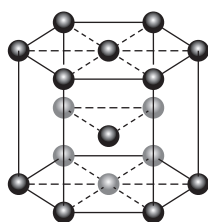
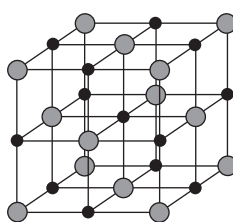
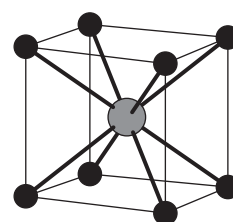
Suggest an equation for the equilibrium formed when hydrazine dissolves in water. [1]

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SECTION BAnswer **all** questions.

8. (a) The diagrams below represent the structures of cadmium metal, caesium chloride, graphite, ice, iodine and sodium chloride.

**A** graphite**B****C****D** cadmium**E****F**

- (i) Label the remaining structures in the spaces provided.

[2]

- (ii) Complete the table showing the trend in melting temperatures of the six substances.

[2]

Substance	Melting temperature
.....	Lowest
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.....	
caesium chloride	
sodium chloride	
.....	
	Highest



- (iii) For structure **B**, name the type of bond or force represented by the letter **X**. [1]

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- (iv) Name the **two** substances which are good electrical conductors in the solid state. [1]

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- (v) Explain why graphite is suitable for use in pencils. [2]

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- (vi) Cadmium is a typical metal.
Give a brief description of metallic bonding. [3]
You may include a diagram to support your answer.

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- (b) Many houses have been built on disused industrial sites. A housing developer wants to test whether the soil on a particular site is contaminated with cadmium ions (Cd^{2+}).

They extract a sample and prepare a solution. Cadmium ions behave in the same way as magnesium ions when treated with sodium hydroxide solution and sodium sulfate solution.

- (i) State what the developer would see if cadmium ions were present when they added:

I. sodium hydroxide solution. [1]

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II. sodium sulfate solution. [1]

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- (ii) Write an ionic equation, including state symbols, for **one** of the above observations. [1]

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8. (a) Aluminium reacts with oxygen to form aluminium oxide.

Using outer electrons only, draw a dot and cross diagram to show the bonding in aluminium oxide.

[2]

- (b) Aluminium also reacts with chlorine to form aluminium chloride.

A 0.400 g sample of aluminium chloride was heated to 220 °C. The vapour produced occupied a volume of 60.8 cm³ at a pressure of 101 kPa.

Show that the molecular formula of aluminium chloride in the vapour is Al₂Cl₆.

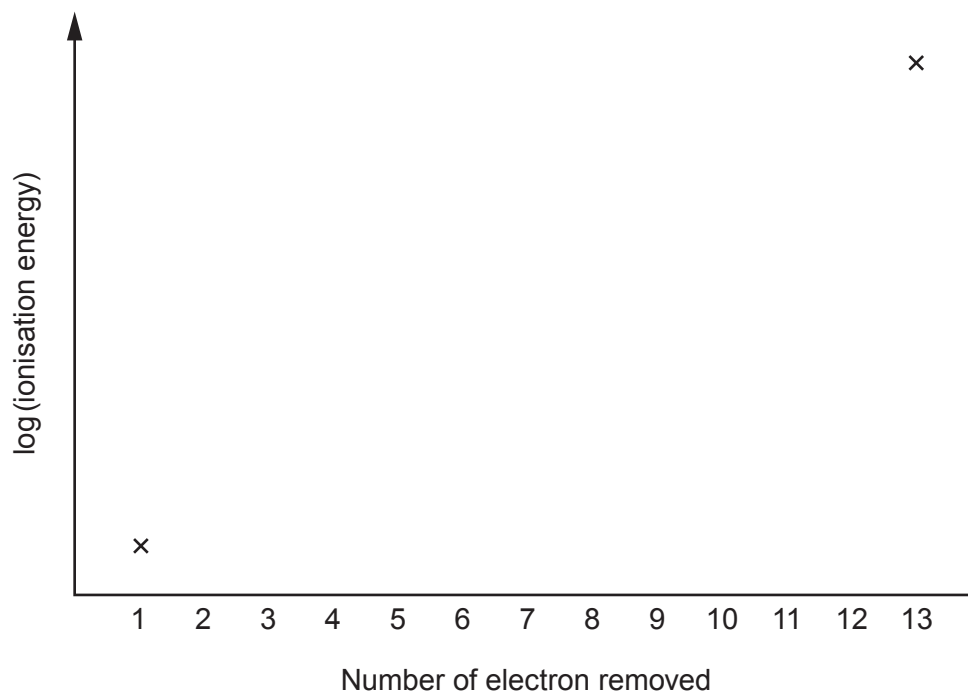
[4]



- (c) (i) Sketch a graph of $\log(\text{ionisation energy})$ for the successive ionisations of aluminium.

The first and last points have been plotted for you.

[1]



- (ii) I. Explain the general slope of the graph.

[1]

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- II. Explain the reason for any sharp changes in the graph.

[1]

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- (d) Although aluminium has 25 known isotopes only two of them occur naturally. These are ^{27}Al which is stable and ^{26}Al which is radioactive.

^{26}Al decays by electron capture and its half-life is 7.2×10^5 years.

- (i) Give the mass number and symbol of the species produced when an atom of ^{26}Al decays. [1]

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- (ii) If 8.0 mg of ^{26}Al decays by electron capture, calculate the mass in **grams** of ^{26}Al left after 2.88×10^6 years. [2]

Mass = g

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